

Communication with Absent Parties in Cybernetics and Adjacent Fields

A Deep Research Synthesis

March 2026

Perplexity Computer

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Executive Summary

What does it mean to communicate when one party is absent, unreachable, or non-responsive? This question — seemingly peripheral to mainstream communication theory, which presupposes co-present, mutually responsive interlocutors — turns out to be a central, cross-cutting problem addressed (often implicitly) across at least eight distinct scholarly traditions. This report synthesizes deep research findings across those eight domains and identifies the absent-party problem as a unifying theoretical thread that connects nuclear semiotics, cybernetic conversation theory, script decipherment, SETI, information cascade dynamics, continental hermeneutics, biosemiotics, and the emerging integrative work of researchers such as Anamaria Berea and Søren Brier.

The central problematic can be stated precisely: in any communicative act, the standard cybernetic model (Wiener, 1948) presupposes a feedback loop — the receiver's response modulates the sender's subsequent output, enabling error correction, mutual calibration, and convergence on shared meaning. When the feedback channel is severed — by deep time (nuclear waste warnings designed to persist 10,000 years), by interstellar distance (SETI), by death (hermeneutic interpretation of historical texts), by radical epistemic incommensurability (undeciphered scripts), by structural decoupling (information cascades), or by biological discontinuity (cross-generational genetic signaling) — the communicative act does not simply degrade. It changes in kind. The sender must anticipate the receiver's entire cognitive and semiotic repertoire without confirmation. The receiver must reconstruct the sender's intentional structure from material residue alone. The message becomes an artifact whose meaning is constituted by the interpretive act, not merely transmitted through it.

Gordon Pask's Conversation Theory (CT) is the cybernetic framework that most explicitly theorizes this boundary: if there is no conversation — no mutual teachback, no agreement over an understanding — then there are, strictly speaking, no CT observables. Yet this boundary is productive, not merely limiting. It defines the space in which feedforward-only communication must operate, and it is precisely the space occupied by nuclear warning design, interstellar messaging, decipherment, and biological heredity.

Across all eight domains surveyed here, we find a recurring structural pattern: the absence of the interlocutor forces the communication system to rely on redundancy (Shannon), multi-modal encoding (Peirce/Jakobson), institutional relay mechanisms (Sebeok), iterability (Derrida), or code-based organic conventions (Barbieri). The theoretical frameworks differ, but the underlying information-theoretic challenge is isomorphic. This report maps these isomorphisms, identifies five major theoretical gaps where the connections remain undertheorized, and proposes Brier's cybersemiotics as a candidate integrative framework.

The eight domains surveyed are: (1) nuclear semiotics and deep-time warning design; (2) Pask's conversation theory and its boundary conditions; (3) script decipherment as inverse hermeneutics; (4) Anamaria Berea's unified research program spanning literary computation, SETI, lunar archives, social media cascades, and biological communication; (5) SETI as

communication theory; (6) information cascades without source feedback; (7) continental hermeneutic traditions from Gadamer through Stiegler; and (8) biosemiotics boundary cases including code biology, epigenetic inheritance, and mycorrhizal networks. A ninth domain — Brier's cybersemiotics — is treated as a candidate integrative framework rather than a separate empirical domain.

Key finding: heredity itself — the transmission of biological information from parent to offspring, where the "sender" (the parental generation) is structurally absent at the moment of "reception" (gene expression in the offspring's developmental context) — is the ur-case of absent-sender communication. Every other domain in this report addresses a specialized variant of the same fundamental problem.

Section 1: The Problem Space — A Taxonomy of Absence

The absent-party communication problem manifests in at least five distinguishable modalities of absence. Each modality imposes different constraints on the communication system and demands different compensatory strategies. The taxonomy below is analytic rather than empirical: real-world cases often combine multiple modalities. Nuclear waste warning, for example, involves temporal, epistemic, and structural absence simultaneously.

1.1 Temporal Absence (Deep Time)

The sender and receiver are separated by spans of time that exceed the lifespan of any known semiotic convention. Languages transform beyond recognition in 1,000–2,000 years; political institutions rarely survive 500 years; material substrates (stone, metal, ceramic) degrade on timescales of 1,000–100,000 years. The paradigm case is nuclear waste repository warning design, where the regulatory period is 10,000 years (WIPP) to 100,000 years (Onkalo). Biological heredity — where genetic information encoded by ancestral organisms is "read" by descendant organisms millions of years later — represents the extreme case of temporal absence.

1.2 Spatial Absence (Interstellar Distance)

The sender and receiver are separated by distances that impose lightspeed-limited delays measured in years, decades, or centuries. The paradigm case is SETI: a signal from a star system 100 light-years away means a 200-year minimum round-trip for any exchange. At interstellar scales, the feedback loop is not merely slow — it is effectively severed, because neither party can assume the other still exists when a response would arrive. Douglas Vakoch's work on active METI (Messaging Extraterrestrial Intelligence) and Freudenthal's Lincos language project both confront this constraint directly.

1.3 Epistemic Absence (Non-Overlapping Frameworks)

The sender and receiver occupy cognitive or semiotic frameworks with no guaranteed overlap. This is the decipherment problem: the receiver possesses a text produced by a sender whose language, conceptual categories, and referential world may share nothing with the receiver's own. Linear B, the Voynich manuscript, and the Indus Valley script each exemplify different degrees of epistemic absence. Gonzalo Munévar's evolutionary epistemology critique of SETI argues that even mathematics — often assumed to be a universal language — may not be shared across independently evolved intelligences.

1.4 Structural Absence (Cascade / Propagation)

The sender's identity, intentions, and context are progressively stripped away as information propagates through a network. In information cascades (Bikhchandani, Hirshleifer, and Welch, 1992), each agent observes the behavior of predecessors and rationally ignores private signals, producing herding effects that are decoupled from the original information source. In Niklas

Luhmann's autopoietic social systems theory, the sender is "structurally in the environment" of the communication system — communication selects information, utterance, and understanding independently of any individual sender's intention. In LLM-generated misinformation, the "sender" may be an algorithmic process with no intentional state at all.

1.5 Biological Absence (Cross-Generational Signaling)

The sender (parental organism or ancestral lineage) is absent at the moment of reception (gene expression, epigenetic activation, niche-constructed environmental modification). Heredity, in the biosemiotic framing of Jesper Hoffmeyer and Marcello Barbieri, is a communication system in which the "message" (the genome, the epigenome, the constructed niche) was composed by an entity that no longer exists and cannot be consulted. Eva Jablonka and Marion Lamb's four-dimensional model of inheritance (genetic, epigenetic, behavioral, symbolic) shows that biological absence is not a single phenomenon but a family of related transmission mechanisms, each with different fidelity, bandwidth, and error-correction properties.

Table 1: Modalities of Absence Mapped to Research Domains

Modality of Absence	Defining Constraint	Research Domains	Paradigm Case
Temporal (Deep Time)	Sender/receiver separated by 10^3 – 10^5 years	Nuclear Semiotics, Biosemiotics	WIPP 10,000-year marker system
Spatial (Interstellar)	Light-speed delay severs feedback loop	SETI, Bera's planetary communication	SETI signal from 100+ light-years
Epistemic (Framework Gap)	No shared code, language, or conceptual schema	Decipherment, SETI, Hermeneutics	Undeciphered scripts; alien signal interpretation
Structural (Cascade)	Source identity stripped by propagation dynamics	Information Cascades, Luhmann's systems theory	Viral misinformation; LLM-generated content
Biological (Cross-Generational)	Sender absent at moment of gene expression	Biosemiotics, Code Biology	Genetic code; epigenetic inheritance

Section 2: Nuclear Semiotics and Long-Term Warning Design

2.1 The WIPP Human Interference Task Force

The field now called nuclear semiotics was formally inaugurated in 1980–1981 when the U.S. Department of Energy and the Bechtel Group commissioned a Human Interference Task Force (HITF) to investigate problems connected with the post-closure marking of a filled nuclear waste repository.¹ The HITF's mandate was specific: devise a method of warning future generations not to mine or drill at the Waste Isolation Pilot Plant (WIPP) near Carlsbad, New Mexico — and to ensure this warning would remain effective for at least 10,000 years, the regulatory period established by the EPA under 40 CFR Part 191.

The key problem articulated by the HITF was not engineering or political but fundamentally communicative: the sender and receiver would be separated not merely by space but by the entirety of possible civilizational change. No shared code could be assumed. No feedback channel would exist. The message had to function regardless of whether human society continued, regressed, advanced, or transformed beyond recognition. The definitive treatment of the resulting marker system proposals appeared in Trauth, Hora, and Guzowski's Sandia report SAND92-1382 (1993), which established the canonical four-level hierarchy of messages, graduated from immediate visceral impact to detailed technical documentation.²

The four levels are: Level 1 (Rudimentary) — "Something man-made is here," communicated through architecture, earthworks, and sound; Level 2 (Cautionary) — "Something man-made is here and it is dangerous," through physical markers and pictographs; Level 3 (Basic) — what, why, when, where, who, and how, through pictographic and multilingual text; and Level 4 (Complex) — detailed technical records, maps, diagrams, and star charts. The paradigmatic Level 1–3 message includes the now-iconic text:

"This place is a message... and part of a system of messages... pay attention to it! This place is not a place of honor... no highly esteemed deed is commemorated here... nothing valued is here. What is here was dangerous and repulsive to us. This message is a warning about danger."

2.2 Sebeok's Atomic Priesthood and the Semiotic Framework

*Thomas A. Sebeok (1920–2001), one of the most distinguished American semioticians, was engaged by Bechtel/ONWI in 1981 as a specialist in signs and communication. His resulting report, **Communication Measures to Bridge Ten Millennia** (BMI/ONWI-532, 1984), is explicitly structured around the Shannon-Weaver communication model as filtered through Roman Jakobson's semiotic extension.³ Sebeok draws directly on Norbert Wiener's distinction between feedback and feedforward: the entire nuclear warning design problem is framed as a pure feedforward problem — the sender must anticipate, without confirmation, the cognitive and semiotic capacities of an unknown receiver.*

Sebeok recommended that effective long-term messages mix all three Peircean sign types: iconic (images of hurt bodies, threat postures), indexical (the site itself as an index of buried danger), and symbolic (text and numerical data). He proposed a relay system dividing the 10,000-year period into segments of approximately 250 years, with living custodians at each relay point translating and re-encoding the warning. His most provocative suggestion was the "atomic priesthood" — a self-perpetuating commission of experts maintaining the truth about repository sites through artificially created ritual and legend, modeled on the Catholic Church's institutional longevity.

The atomic priesthood proposal was ultimately not adopted. Its cultural afterlife includes Sebastian Musch's analysis in *Zygon: Journal of Religion and Science* (2016)⁴ and Marcel Danesi's comprehensive treatment in *Warning Signs: The Semiotics of Danger* (Bloomsbury, 2022), which credits Sebeok as inaugurating the field and extends his framework to danger semiotics broadly.⁵

2.3 Hostile Architecture: Landscape of Thorns

The physical marker proposals generated by the WIPP expert panels represent the most visually striking dimension of nuclear semiotics. Designed primarily by Michael Brill (BOSTI), with drawings by Safdar Abidi, the hostile architecture concepts operated on a semiotic premise: certain forms, shapes, and configurations of landscape communicate danger without depending on learned codes. The "Landscape of Thorns" — a dense, random forest of irregularly sized concrete thorns up to 50 feet high — is paradigmatic. Other designs include the Spike Field, Menacing Earthworks (lightning-bolt-shaped mounds), the Black Hole (an enormous basalt slab rendering land uninhabitable), and Forbidding Blocks (hundreds of house-sized black concrete blocks in an irregular grid with narrow, hot, ominous "streets").⁶

These designs rely on indexical threat (spikes that look like they could wound), shunned-land imagery (landscape that appears already devastated), and the aesthetic of the sublime. The designers were acutely aware of the "treasure-hunter" paradox: anything sufficiently monumental might attract rather than repel. The current DOE marker plan excludes the hostile architecture proposals in favor of berms, granite monuments, buried radar-reflective objects, and multilingual archives.

2.4 Flusser, Post-Literacy, and Technical Images

Vilém Flusser's media theory — particularly his five-stage model of human abstraction (concrete experience → traditional images → linear text → technical images → computation) — addresses precisely the conditions that nuclear warning design must confront.⁷ Flusser's key insight for this problem: the HITF literature generally assumes that future receivers will be **pre-literate** (unable to read modern languages because civilization has collapsed), when the actual danger may be that they will be **post-literate** — able to compute and process images but not trained to decode linear alphabetic text. In either case, the traditional text-based warning fails. Flusser's distinction between dialogical and discursive communication maps directly onto

the feedforward/feedback distinction: nuclear warning is the extreme case of discursive communication, where the sender cannot observe whether the message is received.

2.5 Galison, Onkalo, and Recent Developments

Peter Galison's contribution to nuclear semiotics is primarily one of political epistemology. His essay "Removing Knowledge" (2004) demonstrates that the classified universe of U.S. government documents may be larger than the publicly accessible universe of knowledge.⁸ The paradox Galison illuminates: the same institution (the DOE) that spent decades suppressing public knowledge of nuclear activities is now tasked with ensuring that knowledge survives 10,000 years. His 2015 documentary **Containment** (with Robb Moss) is the most artistically ambitious direct engagement with the WIPP marker problem.⁹

Finland's Onkalo repository represents a radical departure: Posiva Oy has proposed to leave no surface markers when the facility is sealed — a wager that engineered barriers are sufficient and that surface marking, by attracting archaeological attention, might do more harm than good.¹⁰ Michael Madsen's documentary **Into Eternity** (2010) addresses this directly, speaking in second person to an unspecified future viewer and enacting the communication problem it documents.¹¹ Vincent Ialenti's **Deep Time Reckoning** (MIT Press, 2020) provides crucial ethnographic grounding, showing how Finnish nuclear waste experts actually model futures 100,000 years out.¹²

The most significant recent institutional development is the OECD Nuclear Energy Agency's RK&M; Initiative (2011–2018), which compiled a toolbox of 35 mechanisms for record, knowledge, and memory preservation grouped into nine approaches.¹³ Its core principle — that no single mechanism is sufficient and resilience requires diversity — is Shannon-theoretic in principle.

2.6 Jakobson's Communication Model and Its Limits

Roman Jakobson's "Linguistics and Poetics" (1960) — which appeared in Sebeok's edited volume **Style in Language** (MIT Press, 1960) — extended Shannon's model by identifying the functions served by each component of the communication act: referential (context), poetic (message), emotive (addresser), conative (addressee), phatic (contact), and metalingual (code). Sebeok directly invokes Jakobson's taxonomy in framing the nuclear warning problem. The key insight is that nuclear warning design must achieve the conative function (directing future behavior: "do not dig here") in the absence of the phatic and metalingual functions. The phatic function ("is the channel working?") and the metalingual function ("are we using the same code?") both require bidirectional communication. Remove these, and the integrity of the entire communicative act becomes uncertain. This is why pictographic and architectural messages — which operate at the pre-linguistic level — are more promising than text: they bypass the need for shared code.

2.7 The Ray Cat Solution and Biological Channels

Françoise Bastide and Paolo Fabbri, responding to the *Zeitschrift für Semiotik* poll (1982–1984), proposed the "ray cat" solution: genetically engineer cats to change color in the presence of ionizing radiation, coupled with deliberately cultivated folklore explaining that "if your cat changes color, flee." This proposal uses a biological channel rather than a material or institutional one — cats reproduce themselves across generations without human maintenance. It embeds information in the cultural substrate (folklore) rather than in objects that can be destroyed, and it exploits existing human-animal communication codes rather than creating new ones. Stanisław Lem, responding to the same poll, proposed analogous biological channels: "Atomic Flowers" — genetically engineered plants carrying encoded information about waste sites in their DNA. These proposals anticipate the biosemiotic insight (Section 9) that biological systems have evolved the most robust solutions to cross-generational communication.

2.8 Voigt's Concentric Linguistic Rings and Temporal Redundancy

Vilmos Voigt (Eötvös-Loránd University) proposed a graduated translation system: install warning signs in all major global languages in concentric rings around the storage site. As languages change and become unreadable, new signs with translations of the older ones would be added in the next ring out — while the older signs remain. The outer rings would thus provide the key to reading the inner ones, replicating the function of the Rosetta Stone in a planned, deliberate architecture of progressive translation. This is an elegant temporal redundancy solution: rather than designing a single sign system that will survive 10,000 years, design a system that generates its own interpretive key through layered obsolescence. The concept has deep affinities with Shannon's error-correcting codes and with the biological strategy of redundant gene copies (paralogs) that preserve function even as individual copies accumulate mutations.

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Section 3: Pask's Conversation Theory — Boundary Conditions

3.1 CT Fundamentals

Gordon Pask (1928–1996) developed Conversation Theory (CT) over roughly two decades, crystallizing it in three major works: **The Cybernetics of Human Learning and Performance** (Hutchinson, 1975), **Conversation, Cognition and Learning** (Elsevier, 1975), and **Conversation Theory: Applications in Education and Epistemology** (Elsevier, 1976). CT's foundational claim is that knowledge is constituted through conversation — specifically, through a process Pask calls "agreement over an understanding." Two participants (which may be individuals, cognitive sub-systems, or formal agents) engage in mutual teachback: A explains topic T to B, then B demonstrates understanding by teaching T back to A (or by deriving consequences that A recognizes as valid).¹⁴ This mutual demonstration of understanding — not mere signal exchange — is what constitutes a conversation in Pask's sense.

The formal backbone of CT is the entailment mesh: a directed graph of topic nodes connected by entailment relations, where each node represents a procedure for producing or recognizing a concept, and each edge represents a derivation or dependency. Two conversants are said to "agree over an understanding" of topic T when their respective entailment structures for T are isomorphic — not identical, but functionally equivalent in the sense that each can predict the other's derivations.

3.2 Pask on the Limits: No Conversation, No Observables

Pask himself articulated the boundary condition most directly in a passage widely cited by his interpreters: "If there were no conversation, there would be no CT observables." This is not merely a methodological limitation — it is an ontological claim. For Pask, knowledge, concepts, and understanding are not pre-existing entities that conversation reveals; they are constituted by conversation. Without the mutual process of teachback and agreement, there is nothing for CT to describe.

Bernard Scott, Pask's closest collaborator and the principal continuator of CT, has extended this boundary analysis in multiple works.¹⁵ Scott distinguishes between "proto-conversation" (partially formed, asymmetric exchanges that approach but do not fully achieve mutual teachback) and "null conversation" (the complete absence of feedback). The absent-party case falls into the latter category — but Scott recognizes that artifacts produced by a conversant can function as "frozen" entailment structures that a future interpreter can attempt to reconstruct.

3.3 Pangaro's Limit Taxonomy

Paul Pangaro, one of Pask's most influential interpreters and the principal architect of CT's application to interaction design, has developed a taxonomy of communication that explicitly positions Pask's conversational model against Shannon's transmission model.¹⁶ Pangaro

distinguishes: (1) data transfer — Shannon's model, where the receiver merely captures bits; (2) information exchange — the receiver acquires new states; (3) conversation — mutual construction of shared understanding through teachback; and (4) collaboration — joint action emerging from conversational agreement. The absent-party case, in Pangaro's framing, is a case where only level (1) or (2) is available — the message can be transmitted and received, but the conversational closure (level 3) that would constitute genuine understanding cannot occur in real time.

3.4 Laurillard's Absent-Teacher Problem

Diana Laurillard's Conversational Framework — the most widely adopted pedagogical application of Pask's ideas — faces the absent-party problem directly in the context of mediated learning.¹⁷ When a learner interacts with a textbook, a recorded lecture, or a software tutorial, the "teacher" is absent. Laurillard's framework requires adaptation and feedback cycles between teacher and student — cycles that recorded media cannot provide. Her solution is to design media that simulate conversational closure: practice environments with automated feedback, peer discussion prompts, and structured reflection tasks. The design challenge is isomorphic to nuclear semiotics: how do you build a pedagogical artifact that functions in the teacher's absence?

3.5 Green's Axiomatics and Glanville's Second-Order Perspective

Nick Green, building on Pask's later Interaction of Actors Theory (IA), has proposed a set of axioms for interaction that explicitly include conditions for non-conversational communication: amity (willingness to engage), faith (trust that the other is acting in good faith), and resonance (structural compatibility that allows mutual calibration). In the absent-party case, amity and faith must be designed into the artifact itself — the nuclear warning must embody a stance of good-faith communication toward an unknown future receiver.

Ranulph Glanville, Pask's most philosophically oriented student, extended CT into second-order cybernetics by arguing that the observer is always part of the system observed. Glanville's work on "communication without interlocutor" addresses the case where one party in a conversation is absent or replaced by an artifact.¹⁸ His key insight: the absent party's "presence" is constituted by the interpreter's act of interpretation. The conversation does not halt when one party disappears — it becomes a second-order conversation between the interpreter and their own model of the absent other. This reframing turns the absent-party problem from a deficit (missing feedback) into a constitutive condition (the interpreter's creativity is required to complete the circuit).

3.6 Key Finding: The Productive Boundary

The central finding from the CT literature is that CT explicitly defines the absent-party case as its hard boundary — but this boundary is productive, not merely limiting. It defines the space in which feedforward-only communication must operate: the sender must embed enough

structure in the artifact to enable a future interpreter to reconstruct a viable entailment mesh, even without mutual teachback. The artifact becomes a "frozen conversation" — incomplete, but sufficient to seed a new conversational process in the interpreter. This insight connects directly to Eco's "open work," Derrida's iterability, and the biosemiotic concept of the genetic code as a "frozen" message from absent ancestors.

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15. Scott, Pask Archive — <https://www.pangaro.com/Pask-Archive/scott.html>

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17. Laurillard, Teaching as a Design Science — <https://www.routledge.com/Teaching-as-a-Design-Science-Building-Pedagogical-Patterns-for-Learning-and-Technology/Laurillard/p/book/9780415803878>

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Section 4: Decipherment as Inverse Hermeneutics

4.1 The Decipherment Problem as Communication Theory

Decipherment of an unknown writing system is, at its core, a problem of reconstructing the intentional structure of an absent sender from material residue alone. This makes it simultaneously a communication-theoretic problem (a channel with an unknown encoder), an information-theoretic problem (what statistical properties of a signal indicate linguistic versus non-linguistic structure?), and a hermeneutic problem (how does the interpreter's own conceptual structure interact with an alien semiotic system?). The cybernetic tradition — especially second-order cybernetics (von Foerster), conversation theory (Pask), and the ecosystemic epistemology of Bateson — offers conceptual resources that have been only partially applied to decipherment.

4.2 Linear B: Kober's Distributional Analysis as Conditional Entropy

The decipherment of Linear B — achieved by Michael Ventris in 1952, with crucial contributions from Alice Kober and Emmett Bennett — represents the paradigmatic success case. Kober's distributional analysis (performed without computers, using hand-sorted index cards from 1943 to her death in 1950) identified systematic patterns of sign variation that indicated inflectional morphology. In information-theoretic terms, Kober was computing conditional entropy: given a particular sign sequence, what are the probabilities of the signs that follow? Her "triplets" — groups of three related sign sequences sharing a common root but with different endings — revealed the grammatical structure of the language without knowing what any sign meant.¹⁹

The key cybernetic observation: Kober was reconstructing the absent sender's entailment mesh from statistical regularities alone. She did not need to know the meaning of any individual sign; she needed only to detect the structural relationships between signs. This is precisely the procedure that Gregory Bateson characterized as detecting a "difference that makes a difference" — the fundamental unit of information.

4.3 The Voynich Manuscript and Statistical Anomaly

*The Voynich manuscript (Beinecke MS 408) represents the paradigmatic failure case — or, more precisely, the paradigmatic case of underdetermined decipherment. Montemurro and Zanette's 2013 analysis in **PLoS ONE** applied information-theoretic methods to demonstrate that the Voynich text exhibits long-range word-frequency correlations consistent with natural language.²⁰ The text follows Zipf's law, has non-trivial entropy profiles, and shows semantic-level clustering that rules out random generation. Yet no proposed decipherment has achieved scholarly consensus.*

4.4 Computational Approaches: Rao, MIT Neural Decipherment

Rajesh Rao's work on the Indus Valley script applies information-theoretic analysis to determine whether sign sequences exhibit linguistic structure. Rao et al. computed conditional entropy for

Indus sign sequences and found values between those of linguistic and non-linguistic systems, suggesting the Indus signs encode language but with constraints imposed by the medium.²¹ MIT's computational decipherment group (Barzilay, Luo, Knight) has developed neural approaches that treat decipherment as machine translation with a lost source language, successfully applying these to Ugaritic (a known test case) and Linear B before attempting transfer to undeciphered scripts.²²

4.5 The Cybernetic Frame: Eigenforms and Entailment Meshes

The connection to second-order cybernetics is deep but largely implicit in the decipherment literature. Heinz von Foerster's concept of the eigenform — a stable self-referential structure that emerges from recursive operations — maps directly onto the decipherer's process: by iteratively applying hypotheses about sign values and checking for internal consistency, the decipherer converges on a stable interpretation that is both self-consistent and consistent with the evidence. The deciphered text is an eigenform of the interpretive process.

Key gap: No published work explicitly applies Pask's entailment mesh formalism to decipherment. This is a significant theoretical lacuna. The entailment mesh provides a natural formal framework for representing what the decipherer is actually doing: reconstructing the derivational structure (the network of conceptual dependencies) that the absent sender used to produce the text. Kober's triplets, Ventris's grid, and Rao's conditional entropy calculations are all, implicitly, partial reconstructions of an absent sender's entailment mesh.

4.6 Undeciphered Scripts as Limit Cases

Beyond the Voynich manuscript, several historical scripts remain undeciphered, each representing a different configuration of the absent-sender problem. The Indus Valley script (~2600–1900 BCE) presents an extremely short average inscription length (~5 signs), making statistical analysis difficult. Proto-Elamite (~3100–2900 BCE) has approximately 1,600 known tablets with around 1,000 distinct signs — too many for a simple alphabet, suggesting a logographic or logo-syllabic system. The Phaistos Disk (~1700 BCE) is a single artifact with 242 tokens of 45 distinct sign types, printed (not incised) with stamps — suggesting standardized production, but providing too little text for statistical decipherment. Rongorongo from Easter Island (~18th century) presents reverse boustrophedon writing on wooden tablets, most of which were destroyed in the 19th century.

Each undeciphered script instantiates a different degree of epistemic absence. For Linear A (Minoan, ~1800–1450 BCE), the writing system is partially understood (it shares some signs with the deciphered Linear B) but the underlying language is unknown. For the Indus script, even the question of whether the signs encode language at all remains contested (Farmer, Sproat, and Witzel argued in 2004 that the inscriptions are non-linguistic symbols, while Rao's team argued the entropy profiles are linguistic). These cases illustrate that the absent-sender problem has degrees: the sender's "distance" from the interpreter can be measured along multiple dimensions — temporal, linguistic, cultural, and inferential.

4.7 Bateson, Von Foerster, and the Cybernetics of Decipherment

Gregory Bateson's definition of information as "a difference that makes a difference" (from **Steps to an Ecology of Mind**, 1972) provides the minimal cybernetic criterion for detecting communication in an unknown system: if a variation in the signal correlates with a variation in context or behavior, information is present. This is precisely what Kober's distributional analysis detects — systematic differences between sign sequences that correlate with grammatical function. Heinz von Foerster's constructivist epistemology extends this insight: the decipherer does not "discover" the message hidden in the text; rather, the decipherer constructs a model of the absent sender that is consistent with the evidence. The deciphered text is an eigenform of the decipherer's recursive modeling process — a stable self-referential structure that emerges from iterative hypothesis-testing.

This constructivist reframing has important implications. It means that decipherment is not a passive "reading" of a message but an active construction that depends on the decipherer's own cognitive and cultural resources. Different decipherers, with different pre-understandings (in Gadamer's sense), may construct different models of the absent sender — and there may be no fact of the matter about which is "correct" beyond internal consistency and predictive adequacy. The Voynich manuscript may be the case where this ambiguity is irreducible: the evidence may be consistent with multiple incompatible decipherments (or with no decipherment at all, if the text is a hoax).

19. Linear B, Wikipedia — https://en.wikipedia.org/wiki/Linear_B

20. Montemurro & Zanette, PLoS ONE (2013) — <https://doi.org/10.1371/journal.pone.0066344>

21. Rao et al., Science (2009) — <https://doi.org/10.1126/science.1170391>

22. Luo et al., TACL (2019) — https://doi.org/10.1162/tacL_a_00169

Section 5: Anamaria Berea — A Unified Research Program

5.1 Academic Profile

Anamaria Berea is an Associate Term Professor in the Department of Computational and Data Sciences at George Mason University, with affiliations at the SETI Institute, NASA Frontier Development Lab, and the Blue Marble Space Institute of Science.²³ She holds dual PhDs — in Computational Social Science from GMU's Krasnow Institute (2012, the first woman to earn this degree from GMU) and in Economics from the Academy of Economic Studies of Romania (2010). Her self-described research program is explicitly situated at "the intersection of data sciences (with a focus on natural language processing), economics and astrobiology," with the aim of understanding "communication in biological and social networks as an evolving complex system."

5.2 The Folger/Literary Genome Project

During her tenure at ATA, LLC (2016–2018), Berea led a "literary genome" NLP project in collaboration with the Folger Shakespeare Library — the world's largest collection dedicated to Shakespeare.²⁴ The project applied natural language processing, text mining, and semantic analysis to Renaissance and Early Modern texts. In the absent-party framing, this work treats the text as a signal whose latent structure (vocabulary distributions, semantic networks, stylometric patterns) can be reverse-engineered to reconstruct something about authorial intent — the same epistemological challenge as decoding an undeciphered manuscript or interpreting an extraterrestrial transmission.

5.3 The ASPIRE ONE Lunar Archive

The ASPIRE ONE Lunar Record is a deep-time cultural and scientific archive placed on the Moon, designed to preserve Earth's heritage for future intelligent entities. It launched on January 15, 2025 as part of the Ghost Riders in the Sky Lunar Mission via Firefly Aerospace's Blue Ghost Lander and successfully landed on Mare Crisium on March 2, 2025.²⁵ The payload — housed in a LifeShip Pyramid of anodized aluminum — includes a 1GB microchip, a 1MB NanoFiche analog capsule, and synthetic DNA encoding. The content spans a 24-minute "Daily Life" video, constitutions of 192 countries, Apollo 11 launch codes, and works by Newton, Curie, and Asimov.

The theoretical framework is explicitly receiver-agnostic: the archive is designed to be meaningful to "archaeologists or future pioneers" — human, posthuman, or otherwise. Format choice (video blending visuals, subtitles, and code) is itself a semiotic strategy deemed most likely to be decipherable by beings with no shared conventions. Redundancy across media types (digital + analog + DNA) provides degradation tolerance across radically different technological futures.

5.4 Technosignatures and SETI Work

Berea edited **Technosignatures for Detecting Intelligent Life in Our Universe: A Research Companion** (Wiley-Scrivener, 2022), presenting novel ideas from economics, forecasting, and data sciences applied to technosignature research.²⁶ She has co-authored multiple white papers with Douglas Vakoch (METI International), including the NASA Technosignatures 2018 Workshop report.²⁷ Her 2025 Penn State SETI Symposium talk, "Language of Stars," applies information-theoretic methods to categorize communication patterns from impossible to probable, drawing on computational linguistics approaches rooted in Shannon and Kolmogorov.²⁸

5.5 The 'Message in a Bottle' Paper

Jiang, Berea et al. (2023), "Message in a Bottle — An Update to the Golden Record," published in **Earth and Space Science**, directly precedes and informs ASPIRE ONE. The paper explicitly states: "Given the absence of mutually understood signs, symbols, and semiotic conventions, MIAB seeks to use scientific methods to design an innovative means of communication."²⁹ This is a direct engagement with semiotics — specifically the problem of designing messages without shared sign systems.

5.6 'Towards a Generalized Framework for Planetary Communication' (2025)

Berea, Lewis, and Beinhoff's 2025 paper in **Space Policy** develops epistemological, ontological, and methodological frameworks for communication using semantics, pragmatics, and discourse analysis.³⁰ This is the most explicit theoretical articulation of Berea's research program as a unified treatment of the absent-party problem.

5.7 Key Insight: A Unified Program

Berea's entire research program addresses the absent-party problem from five empirical angles: literary texts (Folger), SETI/technosignatures, lunar archives (ASPIRE ONE), social media cascades (DARPA SocialSim), and biological communication (**Emergence of Communication in Socio-Biological Networks**, Springer, 2018). Her theoretical toolkit — information theory, complex systems, agent-based modeling, NLP, Bayesian networks — is applied consistently across all domains. The absent-party communication problem is not a metaphor in her work; it is the literal design challenge that motivates her research agenda.

23. Berea, GMU Faculty Page — <https://science.gmu.edu/directory/anamaria-berea>

24. Berea CV, GMU — <https://science.gmu.edu/media/anamaria-bereacvlongpdf>

25. GMU News, ASPIRE ONE (2025) —

<https://www.gmu.edu/news/2025-01/george-mason-scientist-anamaria-berea-leads-team-design-lunar-cultural-archive>

26. Berea (ed.), *Technosignatures* (2022) — <https://www.scrivenerpublishing.com/cart/title.php?id=704>

27. NASA Technosignatures Workshop Report —

<https://www.hou.usra.edu/meetings/technosignatures2018/agenda/Technosignature-Report.pdf>

28. Penn State SETI Symposium 2025 — <https://sites.psu.edu/setisymposium2025/agenda/talk-abstracts/>

29. Jiang, Berea et al., arXiv (2023) — <https://arxiv.org/abs/2306.01765>

30. Berea et al., *Space Policy* (2025) — <https://www.sciencedirect.com/science/article/abs/pii/S0265964625000359>

Section 6: SETI as Communication Theory

6.1 Vakoch's Exosemiotics

Douglas Vakoch, president of METI International, is the scholar who has most explicitly positioned SETI as a problem in communication theory rather than merely signal detection. His edited volumes — **Archaeology, Anthropology, and Interstellar Communication** (NASA SP-2013-4413, 2014), **Communication with Extraterrestrial Intelligence** (SUNY Press, 2011), and **The Drake Equation: Estimating the Prevalence of Extraterrestrial Life through the Ages** (Cambridge, 2015) — systematically bridge SETI with semiotics, hermeneutics, and communication theory.³¹ Vakoch's concept of exosemiotics — the study of sign systems that might be used in interstellar communication — confronts the absent-party problem in its most radical form: the sender and receiver may share no evolutionary history, no perceptual apparatus, no cognitive architecture, and no semiotic conventions.

6.2 Munévar's Evolutionary Epistemology Critique

Gonzalo Munévar's philosophical critique of SETI argues that the assumption of universal mathematical communication is unfounded. Drawing on evolutionary epistemology, Munévar contends that mathematical structures may be contingent on the specific evolutionary and developmental history of the species that produces them. An independently evolved intelligence might have cognitive structures for which our mathematics is simply not a natural description.³² This is a radical deepening of the epistemic absence modality: if even mathematics is not universal, the feedback-free communication problem becomes genuinely intractable.

6.3 Freudenthal's Lincos and Constructed Languages

Hans Freudenthal's **Lincos: Design of a Language for Cosmic Intercourse** (1960) was the first systematic attempt to construct a language for interstellar communication from mathematical primitives.³³ Lincos begins with natural numbers and arithmetic, builds up to logical connectives, variables, quantifiers, time, behavior, space, and eventually social concepts. Later extensions include Carl DeVito and Richard Oehrle's revised Lincos (1990s) and Yvan Dutil and Stéphane Dumas's "Cosmic Call" messages (1999, 2003). The fundamental assumption is that mathematics provides a shared semiotic ground — precisely the assumption Munévar challenges.

6.4 Hermeneutic SETI: Dubinina and Tsybulco

Dubinina and Tsybulco's 2019 work applies Gadamerian hermeneutics to SETI, arguing that the "fusion of horizons" between human and extraterrestrial sign systems is the central challenge.³⁴ Their "hermeneutic SETI" approach treats an intercepted alien signal not as data to be decoded but as a text to be interpreted — with all the hermeneutic circularity that implies. The interpreter must bring a pre-understanding to the signal, which is then modified by the signal's resistance to that pre-understanding. This is Gadamer's hermeneutic circle applied to the most extreme

case of absent-sender communication.

6.5 Doyle's Information Theory and Animal Communication

Laurance Doyle's information-theoretic analysis of animal communication systems provides the empirical bridge between SETI and biosemiotics. Doyle, working at the SETI Institute with collaborators Brenda McCowan and Sean Hanser, applied Zipf's law and Shannon entropy to humpback whale songs and bottlenose dolphin vocalizations, demonstrating that these systems exhibit statistical structure consistent with rule-governed communication.³⁵ The methodological insight is profound: the same information-theoretic tools used to assess whether a SETI signal is linguistic can be applied to non-human communication on Earth. Animal communication is the intermediate case between human and extraterrestrial — the sender is present but does not share our semiotic framework.

6.6 Denise Herzing and the Wild Dolphin Project

Denise Herzing's three-decade Wild Dolphin Project and the CHAT (Cetacean Hearing and Telemetry) system represent the most sustained empirical effort to establish two-way communication with a non-human species.³⁶ Herzing's work is significant for the absent-party problem because it tests, *in vivo*, the conditions under which communication can be established between cognitive systems with non-overlapping semiotic frameworks — the terrestrial analog of SETI's interstellar challenge.

6.7 Zoo Hypothesis and Dark Forest as Communication Theories

The Zoo Hypothesis (Ball, 1973) and Liu Cixin's Dark Forest theory (from **The Three-Body Problem**, 2008/English 2014) are not merely speculative solutions to the Fermi Paradox — they are theories about the communication dynamics of absent parties. The Zoo Hypothesis posits that extraterrestrial civilizations are present but deliberately non-communicative — structurally absent by choice. The Dark Forest theory posits that civilizations are absent from the communication channel because visibility is lethal: any civilization that reveals its position is destroyed by others. Both theories frame absence as a strategic communication decision rather than a physical constraint.

6.8 Key Finding

SETI is the discipline that has most explicitly theorized the absent-party problem. The SETI Post-Detection Hub, established in 2023 at the University of St Andrews under John Elliott, coordinates international preparation for how to interpret and respond to a confirmed extraterrestrial signal.³⁷ SETI's central contribution to the broader synthesis is the recognition that communication without feedback is not an edge case — it may be the normal condition of communication across cosmic scales.

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31. Vakoch (ed.), NASA SP-2013-4413 (2014) — https://www.nasa.gov/wp-content/uploads/2015/01/archaeology_anthropology_and_interstellar_communication_tagged.pdf
 32. Munévar in Vakoch (ed.), *Communication with ETI* — https://link.springer.com/chapter/10.1007/978-3-642-27437-4_8
 33. Lincos, Wikipedia — [https://en.wikipedia.org/wiki/Lincos_\(artificial_language\)](https://en.wikipedia.org/wiki/Lincos_(artificial_language))
 34. Dubinina & Tsybulco, *Acta Astronautica* (2019) — <https://doi.org/10.1016/j.actaastro.2019.01.002>
 35. Doyle et al., *Acta Astronautica* (2011) — <https://doi.org/10.1016/j.actaastro.2011.05.014>
 36. Wild Dolphin Project — <https://www.wilddolphinproject.org/>
 37. SETI Post-Detection Hub — <https://seti-post-detection.org/>

Section 7: Information Cascades Without Source Feedback

7.1 DARPA SocialSim

The Computational Simulation of Online Social Behavior (SocialSim) program was launched by DARPA to develop "innovative technologies for high-fidelity computational simulation of online social behavior" focused on information spread and evolution. The program funded multiple performers, with UCF's Complex Adaptive Systems Laboratory's Deep Agent system (led by Ivan Garibay, with Berea as scientific lead, total funding approximately \$6.2 million) being a major performer.³⁸ Deep Agent used agent-based models and deep reinforcement learning to simulate cross-platform information cascades across Reddit, Twitter, and GitHub.

7.2 Foundational Cascade Theory: BHW 1992

*Bikhchandani, Hirshleifer, and Welch's 1992 paper "A Theory of Fads, Fashion, Custom, and Cultural Change as Informational Cascades" in the **Journal of Political Economy** established the foundational model of information cascades.³⁹ The BHW model describes a sequential decision problem in which each agent observes the actions (but not private signals) of predecessors and rationally updates beliefs. Beyond a critical mass, agents rationally ignore their private information and follow the herd — producing a cascade. The connection to absent-party communication is structural: as the cascade propagates, the identity, intentions, and evidential basis of the original information source are progressively stripped away. The "sender" becomes absent not through distance or death but through the dynamics of propagation itself.*

7.3 Daley–Kendall Rumor Dynamics

The Daley–Kendall model (1965) formalizes rumor propagation as an epidemiological process with three population classes: ignorants (susceptibles), spreaders (infectives), and stiflers (removed). The model captures a key feature of absent-sender communication: the rumor's content and framing evolve as it passes through the population, with no mechanism for the original sender to correct drift or distortion. Each retelling is a lossy re-encoding with no error-correction channel back to the source.

7.4 Vosoughi, Roy, and Aral (2018): The Spread of True and False News

*Vosoughi, Roy, and Aral's 2018 **Science** study of 126,000 rumor cascades on Twitter (now X) found that false news spreads "farther, faster, deeper, and more broadly than the truth in all categories of information."⁴⁰ The effect was driven primarily by human behavior (novelty, emotional valence) rather than bots. This finding has direct implications for absent-sender communication: the original sender's intended meaning is not merely diluted by propagation — it is systematically distorted by the receiver population's cognitive biases.*

7.5 Wardle and Derakhshan (2017): Information Disorder Framework

Claire Wardle and Hossein Derakhshan's 2017 Council of Europe report "Information Disorder: Toward an Interdisciplinary Framework for Research and Policy Making" introduced a tripartite taxonomy: mis-information (false, shared without intent to harm), dis-information (false, shared with intent to harm), and mal-information (true, shared with intent to harm).⁴¹ The absent-sender framing unifies all three: once information enters the cascade, the sender's intent (or lack thereof) is irrelevant to propagation dynamics. The information behaves as an autonomous agent in the network.

7.6 Luhmann's Autopoietic Communication

Niklas Luhmann's three-selection model of communication (information, utterance, understanding) provides the deepest theoretical reframing: in Luhmann's systems theory, the sender is structurally "in the environment" of the communication system. Communication is an operation of social systems, not of individuals. The individual who "sends" a message is not the communication's author in any strong sense — the communication system selects what counts as information, what counts as utterance, and what counts as understanding, according to its own operational logic. In this framework, the absent sender is not an anomaly but the normal condition: all communication is, in a precise sense, communication without a sender.

7.7 LLM Misinformation as New Absent-Sender Form

The emergence of large language models (2022–2026) has created an entirely new category of absent-sender communication: algorithmically generated text with no intentional agent behind it. LLM-generated misinformation is "absent-sender" in the strongest possible sense — there is no sender, only a stochastic process that produces plausible-looking text. This collapses the distinction between mis-information and dis-information: the "intent" dimension of Wardle and Derakhshan's framework becomes inapplicable when the source is a language model. The cascade dynamics of LLM-generated content are a natural extension of the BHW framework, with the "original information source" replaced by a process that has no private signal at all.

38. DARPA SocialSim Program — <https://www.darpa.mil/program/computational-simulation-of-online-social-behavior>

39. Bikhchandani et al., JPE (1992) — <https://doi.org/10.1086/261849>

40. Vosoughi et al., Science (2018) — <https://doi.org/10.1126/science.aap9559>

41. Wardle & Derakhshan, Council of Europe (2017) — <https://rm.coe.int/information-disorder-toward-an-interdisciplinary-framework-for-research/168076277c>

Section 8: Adjacent Hermeneutic Traditions

8.1 Gadamer's Frozen Horizon

Hans-Georg Gadamer's philosophical hermeneutics, developed primarily in **Truth and Method** (1960), centers on the concept of *Horizontverschmelzung* — the "fusion of horizons." A "horizon" is the range of vision from a particular vantage point: the interpreter approaches a text with their own horizon of expectations, prejudices, and questions, while the text embodies the horizon of its absent author. Understanding occurs when these horizons fuse — when the interpreter's pre-understanding is modified by the text's resistance to it, producing a new horizon that encompasses both.

For the absent-party problem, the critical concept is the frozen horizon: when the author is dead or otherwise unreachable, their horizon is fixed — it cannot expand, adapt, or respond to the interpreter's questions. All the work of bridging must be done by the interpreter. The fusion is asymmetric: the interpreter fuses with a static structure, like a dancer partnering a statue. This asymmetry is precisely analogous to the feedforward condition in nuclear semiotics: the sender cannot adjust.

8.2 Derrida: Iterability as Constitutive Absence

Jacques Derrida's "Signature Event Context" (1972) provides the most radical philosophical treatment of the absent-party problem. Derrida argues that iterability — the capacity of a sign to be repeated in the absence of its original sender, receiver, and context — is not an accident that befalls communication but its constitutive condition.⁴² A sign that could only function in the presence of its author would not be a sign at all. Writing, in Derrida's analysis, is defined precisely by its capacity to function when the author is absent, dead, or unknown. The "absence of the sender" is therefore not a special case — it is the general condition of all communication, merely foregrounded in certain extreme cases (nuclear warnings, interstellar messages, ancient texts).

8.3 Ricoeur's Distanciation

Paul Ricoeur's hermeneutic theory centers on distanciation — the process by which a text becomes autonomous from its author's intention and original context. Once inscribed, the text "says" something that the author may not have intended, and it addresses readers the author never imagined. Ricoeur distinguishes the text's "sense" (what it says, its internal semantic structure) from its "reference" (what it is about, its relation to the world). Distanciation liberates the text from the author's psychology and makes it available for new interpretive acts — but it also means the author cannot correct misreadings. For the absent-party problem, distanciation is not a loss but a precondition for cross-temporal communication.

8.4 Eco: Open Work and Model Reader

Umberto Eco's theory of the "open work" (***Opera Aperta***, 1962) and his concept of the "Model Reader" provide a semiotic framework for understanding how texts designed for absent receivers function. The open work is structured to invite multiple interpretations without collapsing into arbitrary meaning — it provides enough constraint to guide reading while leaving enough openness to accommodate readers with different horizons.⁴³ Eco's "Model Reader" is the hypothetical ideal interpreter inscribed in the text's structure — the reader the text "wants." Nuclear warning design, SETI messaging, and archival communication all require constructing a Model Reader for an entity that does not yet exist.

8.5 Hayles: Bridging Cybernetics and Literary Theory

N. Katherine Hayles's ***How We Became Posthuman*** (1999) is the most sustained attempt to bridge cybernetics, information theory, and literary/cultural theory. Hayles traces how the cybernetic concept of information was progressively disembodied — separated from the material substrate that carries it — and argues that this disembodiment has profound consequences for how we understand communication with absent parties.⁴⁴ If information is substrate-independent (as Shannon's theory implies), then it can survive the destruction of its original medium — but Hayles argues that meaning is always embodied, so the substrate matters. The nuclear warning problem illustrates Hayles's point: the same information encoded in stone, in language, in architecture, and in DNA is not the "same" message, because each medium shapes reception.

8.6 Luhmann's Three-Selection Model

As discussed in Section 7, Luhmann's communication theory treats the sender as structurally external to the communication system. Communication is not the transfer of meaning from one mind to another but a three-part selection: information (what is communicated), utterance (how it is communicated), and understanding (how it is received). Each selection is an operation of the social system, not of individual psychology. This framework makes the absent sender not an anomaly but the ordinary condition — and it provides the most rigorous sociological account of why communication continues to function when the sender is gone.

8.7 Kittler: Post-Hermeneutic Media Theory

Friedrich Kittler's media theory proposes that media technologies — not human intentions — determine what can be communicated. In ***Discourse Networks 1800/1900*** (1985) and ***Gramophone, Film, Typewriter*** (1986), Kittler argues that each media epoch creates its own conditions of possibility for communication, and that meaning is an effect of the medium rather than an intention of the author.⁴⁵ For the absent-party problem, Kittler's post-hermeneutic stance suggests that the question is not "what did the sender mean?" but "what does the medium permit?" — a radical reframing that shifts attention from the absent sender to the material infrastructure of transmission.

8.8 Stiegler: Tertiary Retention

Bernard Stiegler's concept of tertiary retention — drawing on Husserl's phenomenology of time-consciousness — provides the most direct philosophical account of how artifacts carry meaning across generations. Primary retention is the just-past moment still held in consciousness; secondary retention is memory; tertiary retention is externalized memory — tools, texts, recordings, institutions. Stiegler argues that tertiary retention is constitutive of human temporality: we are the beings who inherit and transmit through artifacts.⁴⁶ Nuclear warning markers, lunar archives, genetic codes, and LLM-generated texts are all forms of tertiary retention — messages from absent senders preserved in material substrates.

42. Derrida, *Signature Event Context* (1972) — <https://doi.org/10.2307/1343073>

43. Eco, *The Open Work* — https://en.wikipedia.org/wiki/The_Open_Work

44. Hayles, *How We Became Posthuman* (1999) — <https://press.uchicago.edu/ucp/books/book/chicago/H/bo3769963.html>

45. Kittler on Monoskop — https://monoskop.org/Friedrich_Kittler

46. Stiegler, Wikipedia — https://en.wikipedia.org/wiki/Bernard_Stiegler

Section 9: Biosemiotics Boundary Cases

9.1 Barbieri's Code Biology and Organic Codes

Marcello Barbieri (b. 1943, Ferrara) is the central figure in code biology, a program he distinguishes sharply from mainstream biosemiotics. His career-defining insight is that biological life is constituted not by copying alone but by coding — and that codes are irreducibly different from physical laws or chemical reactions. An organic code requires two independent molecular "worlds" (e.g., nucleotides and amino acids), a set of adaptor molecules that create a mapping between them (e.g., tRNAs), and demonstration that the mapping is arbitrary.⁴⁷ The genetic code satisfies all three criteria: tRNAs map codons to amino acids, and the mapping is not chemically determined. Barbieri identifies at least 280 organic codes in biology, from the histone code to the signal transduction code, each connecting distinct molecular domains through arbitrary but stable mappings.

For the absent-party problem, Barbieri's contribution is foundational: the genetic code is a message from absent ancestors. The "sender" (the ancestral organism that established the code) is not present when the code is "read" (during translation in the ribosome). The code's meaning is constituted by the molecular machinery that interprets it, not by the sender's intention. This is the biological analog of Derrida's iterability: the genetic code functions precisely because it can operate in the sender's absence.

9.2 Hoffmeyer's Semiosphere and Semiotic Freedom

Jesper Hoffmeyer (1942–2019), the leading figure of the Copenhagen school of biosemiotics, introduced the concept of the semiosphere — the sphere of sign processes that pervades and constitutes the living world, analogous to Vernadsky's biosphere.⁴⁸ Hoffmeyer's concept of semiotic freedom — the capacity of organisms to interpret signs in multiple ways, increasing with evolutionary complexity — provides a biosemiotic measure of how much "interpretive latitude" a receiver has. Higher semiotic freedom means a greater capacity to extract meaning from ambiguous or degraded signals — precisely the capacity needed to interpret messages from absent senders.

9.3 Pattee's Epistemic Cut

Howard Pattee's concept of the epistemic cut — the irreducible distinction between the physical dynamics of a system and its symbolic description — is fundamental to understanding how biological communication works across generations. The genetic code operates at the epistemic cut: DNA is simultaneously a physical molecule subject to chemical laws and a symbolic structure subject to semiotic interpretation.⁴⁹ Pattee argues that this duality is irreducible — neither the physical nor the symbolic description can be derived from the other. For the absent-party problem, the epistemic cut explains why biological messages survive the sender's death: the symbolic dimension is substrate-independent enough to be read by a new interpreter, while the physical dimension ensures material continuity.

9.4 Epigenetic Inheritance as Cross-Temporal Messaging

Eva Jablonka and Marion Lamb's four-dimensional model of inheritance (genetic, epigenetic, behavioral, symbolic) demonstrates that biological communication across generations operates through multiple channels with different fidelity, bandwidth, and error-correction properties.⁵⁰ Epigenetic inheritance — the transmission of gene-expression patterns through DNA methylation, histone modification, and small RNA pathways — is particularly significant for the absent-party framing: the "message" (the epigenetic mark) was set by environmental conditions experienced by the parent, but it is "read" by the offspring in a potentially different environment. The sender (the parental generation's cellular machinery) is absent, and the receiver (the offspring's developmental process) must interpret the mark without access to the sender's context.

9.5 Niche Construction and Mycorrhizal Networks

Odling-Smee, Laland, and Feldman's niche construction theory (2003) extends the absent-sender framework to ecological inheritance: organisms modify their environments in ways that affect the selection pressures experienced by future generations.⁵¹ The beaver dam, the earthworm's soil modification, and the human-built environment are all messages to future organisms — feedforward communications embedded in the physical world rather than in a symbolic medium. Mycorrhizal networks — the underground fungal networks connecting trees in forests — represent a particularly striking case: they mediate chemical signaling between plants that may be separated by meters or decades, creating a communication system where the "sender" may no longer exist when the signal is received.

9.6 Deacon's Absential Features

Terrence Deacon's ***Incomplete Nature: How Mind Emerged from Matter*** (2012) introduces the concept of absential features — properties defined by what is absent rather than what is present. Information, function, purpose, and meaning are all absential: they are defined by the constraints they impose, the possibilities they exclude, the states they prevent.⁵² Deacon's framework provides a deep theoretical justification for treating absence as constitutive rather than privative. The absent sender is not a deficiency in the communication system — the sender's absence is what makes the message a message rather than a direct interaction.

9.7 Kull and the Tartu School

Kalevi Kull, the leading figure of the Tartu school of biosemiotics (following Jakob von Uexküll's tradition), emphasizes the *Umwelt* — the organism's subjective, species-specific perceptual world — as the fundamental unit of biosemiotic analysis.⁵³ Communication between organisms with different *Umwelten* is communication across an epistemic gap — a biological analog of the SETI problem. The Tartu school's contribution is to show that this gap is not the exception but the rule: every act of inter-species communication (and, by extension, every act of cross-generational communication) involves interpretation across incommensurable

experiential worlds.

9.8 Key Insight: Heredity as the Ur-Case

The central insight from the biosemiotics literature is that heredity itself is the ur-case of absent-sender communication. The genetic code, epigenetic marks, niche modifications, and behavioral traditions are all messages from ancestors who are no longer present. The biological world has been solving the absent-sender problem for 3.8 billion years — and the solutions it has evolved (redundant coding, modular architecture, error-correction mechanisms, layered transmission channels) are strikingly convergent with the engineering solutions proposed for nuclear warning design and interstellar messaging.

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- 47. Barbieri, *Biosemiotics* (2015) — <https://doi.org/10.1007/s12304-014-9236-4>
 - 48. Hoffmeyer, *Semiotic Freedom* — https://doi.org/10.1007/978-94-007-4917-0_11
 - 49. Pattee, *BioSystems* (2001) — [https://doi.org/10.1016/S0303-2647\(00\)00110-1](https://doi.org/10.1016/S0303-2647(00)00110-1)
 - 50. Jablonka & Lamb, *Evolution in Four Dimensions* (2014) — <https://mitpress.mit.edu/9780262525848/evolution-in-four-dimensions/>
 - 51. Odling-Smee et al., *Niche Construction* (2003) — <https://press.princeton.edu/books/paperback/9780691044378/niche-construction>
 - 52. Deacon, *Incomplete Nature* (2012) — <https://www.norton.com/books/Incomplete-Nature/>
 - 53. Kull, *Chinese Semiotic Studies* (2018) — <https://doi.org/10.1515/css-2018-0003>

Section 10: Brier's Cybersemiotics — The Integrative Framework

10.1 The Problem of Integration

The eight domains surveyed in this report draw on distinct theoretical traditions — Shannon's information theory, Pask's conversation theory, Peircean semiotics, Gadamerian hermeneutics, Luhmann's autopoiesis, Barbieri's code biology — that have developed largely in isolation. Søren Brier's cybersemiotics represents the most systematic attempt to bridge these traditions into a single coherent framework.

10.2 Brier's Framework

*Brier's foundational work, **Cybersemiotics: Why Information Is Not Enough** (University of Toronto Press, 2008), argues that neither information theory nor semiotics alone can provide a complete account of communication and cognition.⁵⁴ Information theory (Shannon, Wiener) captures the syntactic dimension — the structure and transmission properties of signals — but cannot account for meaning, interpretation, or lived experience. Semiotics (Peirce, Sebeok) captures the semantic and pragmatic dimensions — how signs mean and how they are used — but lacks the mathematical rigor and the account of physical substrate that information theory provides.*

Brier proposes a five-level model integrating these traditions:

Level 1: Physical-chemical information processing. The domain of Shannon's mathematical theory and thermodynamic entropy. Signs at this level are physical configurations subject to natural laws.

Level 2: Biological autopoiesis and code biology. The domain of Maturana and Varela's autopoiesis and Barbieri's organic codes. Living systems process information through self-referential, organizationally closed processes.

Level 3: Psychological-phenomenological experience. The domain of Peirce's firstness — qualia, feelings, the "inner" dimension of sign processes. This is the level that pure information theory cannot reach.

Level 4: Social communication systems. The domain of Luhmann's autopoietic social systems and Habermas's communicative action. Communication is an operation of social systems that cannot be reduced to individual psychology.

Level 5: Linguistic-conceptual frameworks. The domain of Pask's conversation theory and Gadamer's hermeneutics. Understanding is constituted through mutual interpretation and the fusion of horizons.

10.3 Cybersemiotics and the Absent-Party Problem

Brier's framework could potentially unify the eight domains surveyed here by providing a consistent account of how absence operates at each level. At Level 1, the absent sender's signal persists as a physical trace (the nuclear marker, the DNA sequence). At Level 2, the biological interpreter's autopoietic processes determine how the trace is "read." At Level 3, the phenomenological experience of encountering the absent sender's message shapes interpretation. At Level 4, social systems process the message according to their own operational logic. At Level 5, the interpreter's entailment mesh engages with the frozen entailment mesh of the absent sender.

However, Brier's work does not explicitly address the absent-party problem in these terms. His focus is on the integration of existing theoretical traditions rather than on the specific case of communication without feedback. The application of cybersemiotics to the absent-party problem remains an open research program. Recent work by Brier in the **Journal of Biosemiotics** and through the International Association for Cybersemiotics and the journal **Cybernetics & Human Knowing** continues to develop the framework.⁵⁵

10.4 Brier's Critique of Computational Information Theory

A central argument in Brier's work is that the Shannon-Wiener information concept is necessary but not sufficient for understanding communication. Shannon's information is purely syntactic — it measures the reduction of uncertainty in a signal but says nothing about what the signal means or how it is experienced. Wiener's cybernetic feedback model adds the dimension of control and self-regulation, but it remains agnostic about subjective experience. Brier argues that Peirce's triadic semiotics (sign, object, interpretant) provides the missing dimension: the interpretant — the effect the sign produces in the interpreter — is irreducibly experiential and cannot be captured by a purely formal or computational description.

For the absent-party problem, this critique is consequential. If the Shannon-Wiener framework were sufficient, absent-party communication would be a purely technical problem of signal preservation and noise reduction — and it would have been solved by the nuclear engineers and information theorists long ago. The fact that it remains unsolved after four decades of effort (the HITF was commissioned in 1981) suggests that something beyond syntactic information preservation is required: what Brier calls the "semantic and pragmatic closure" of the communication act, which can only be achieved through an interpretive process that involves lived experience, cultural context, and semiotic competence.

10.5 Peirce's Semiosis and the Absent Interpretant

Brier's reliance on Peircean semiotics highlights a feature of the absent-party problem that other frameworks obscure: in Peirce's triadic model, every sign relation involves a sign (representamen), an object (what the sign refers to), and an interpretant (the effect produced in the interpreter). The interpretant is itself a sign, producing further interpretants in an infinite chain of semiosis. When the sender is absent, the interpretant chain that the sender would have produced is truncated — the sender's intended interpretant is unavailable. The receiver must

generate their own interpretant chain, which may diverge from the sender's intended chain. This is not a failure of communication but the normal condition of semiosis: Peirce himself recognized that sign interpretation is always fallible and revisable.

The cybersemiotic integration thus provides a principled account of why absent-party communication is genuinely difficult (it requires the receiver to generate an interpretant chain without access to the sender's intended interpretants) while also explaining why it is not impossible (semiosis is inherently open-ended and does not require the sender's presence to proceed). The absent sender's message provides the representamen; the world provides the object; the receiver's semiotic competence generates the interpretant. The circuit is incomplete but functional.

54. Brier, *Cybersemiotics* (2008) — <https://utorontopress.com/9780802092519/cybersemiotics/>

55. *Cybernetics & Human Knowing* — <https://chfrp.wordpress.com/cybernetics-human-knowing/>

Section 11: Cross-Cutting Synthesis

11.1 Comparative Analysis

The following table maps each domain against its modality of absence, the status of its feedback channel, the primary theoretical framework it employs, and its key unsolved problem.

Domain	Type of Absence	Feedback Channel	Framework	Key Unsolved Problem
Nuclear Semiotics	Temporal (10^4 – 10^5 years)	Severed (pure feedforward)	Shannon/Jakobson/ Peirce	No cybernetic framing; no receiver model
Pask's CT	Boundary condition	Absent (no teachback)	Conversation Theory	Productive boundary undertheorized
Decipherment	Epistemic (dead sender)	One-way (text only)	Information theory/ hermeneutics	No Paskian entailment mesh treatment
Berea's Program	Multiple (all five types)	Variable by sub-domain	Complex systems/ NLP/info. theory	Needs 2nd-order cybernetic grounding
SETI	Spatial + Epistemic	Severed (lightspeed delay)	Exosemiotics/ hermeneutics	No confirmed signal; no empirical test
Information Cascades	Structural (propagation)	Stripped by cascade dynamics	Network science/ Luhmann	Bridge to biosemiotics incomplete
Hermeneutics	Temporal + Epistemic	One-way (text → reader)	Gadamer/Derrida/ Ricoeur	Derrida–cybernetics connection weak
Biosemiotics	Biological (cross-gen.)	Absent (ancestor dead)	Code biology/ Peirce	Integration with cascade theory

11.2 Five Major Theoretical Gaps

Gap 1: No Paskian Treatment of Absent-Party Decipherment. Pask's entailment mesh provides a natural formal framework for what decipherers actually do — reconstruct the derivational structure of an absent sender — yet no published work applies this formalism to script decipherment. Kober's triplets, Ventris's grid, and computational approaches are all implicitly entailment-mesh reconstructions, but the connection has never been made explicit.

Gap 2: No Systematic Cybernetic Framing of Nuclear Semiotics. Although Sebeok (1984) draws on Shannon and Wiener, and Flusser's media theory is cybernetics-adjacent, no published work provides a systematic second-order cybernetic analysis of the nuclear warning problem. The HITF literature treats the problem as engineering and semiotics; it does not employ the conceptual vocabulary of von Foerster's observing systems, Maturana's structural coupling, or Pask's conversation theory.

Gap 3: Incomplete Bridge Between Biosemiotics and Information Cascade Theory. Both domains deal with communication where the original source is structurally absent, but the theoretical vocabularies are disjoint. Biosemiotics (Barbieri, Hoffmeyer) uses Peircean sign theory; cascade theory (BHW, network science) uses game theory and epidemiology. Luhmann's autopoiesis could bridge the two, but the connection has not been systematically developed.

Gap 4: Berea's Work Needs Explicit Grounding in Second-Order Cybernetics. Berea's research program is implicitly cybernetic — she studies communication as a complex adaptive system using agent-based models, network analysis, and information theory. But her published work does not explicitly engage with the cybernetic tradition (Wiener, Ashby, von Foerster, Pask, Maturana). Grounding her framework in second-order cybernetics would strengthen its theoretical foundations and connect it to the broader absent-party synthesis.

Gap 5: The Derrida–Cybernetics Connection Remains Undertheorized Despite Stiegler. Derrida's iterability thesis and cybernetic feedback theory address the same structural problem — how communication functions in the absence of presence — but from radically different intellectual traditions. Stiegler's concept of tertiary retention provides a partial bridge, and Hayles's work on cybernetics and literary theory gestures toward the connection, but no published work has systematically mapped the isomorphism between Derrida's constitutive absence and the cybernetic concept of feedforward-only communication.

11.3 Convergent Solutions

Despite the diversity of theoretical frameworks, the eight domains converge on a remarkably consistent set of design principles for absent-party communication:

Redundancy. Shannon's fundamental insight — that redundancy counteracts noise — appears in every domain: the WIPP four-level message system, the multi-modal encoding of ASPIRE ONE (digital + analog + DNA), the genetic code's degeneracy (multiple codons mapping to the same amino acid), the hermeneutic tradition's emphasis on multiple readings, and Eco's open work.

Multi-modal encoding. Peirce's triadic sign theory — iconic, indexical, symbolic — provides the framework: effective absent-party communication uses all three modes simultaneously, so that degradation of any one does not destroy the message.

Institutional relay. Sebeok's relay system, Stiegler's tertiary retention, and the biosemiotic concept of cultural niche construction all involve embedding the message in self-perpetuating social or biological structures that actively maintain and re-encode it across generations.

Designed interpretive latitude. Eco's Model Reader, Hoffmeyer's semiotic freedom, and Laurillard's pedagogical design all address the same challenge: creating messages that are structured enough to guide interpretation but open enough to accommodate receivers with different frameworks.

Eigenform convergence. Von Foerster's eigenforms, Pask's agreement over an understanding, and the decipherer's iterative hypothesis-testing all describe the same process: a self-stabilizing interpretive cycle that converges on a stable meaning without requiring the sender's confirmation.

11.4 The Information-Theoretic Isomorphism

At the deepest level, all eight domains share a common information-theoretic structure. In each case, the communication system must: (1) encode information in a form that survives degradation of the channel (noise); (2) provide sufficient redundancy that the message can be reconstructed from partial reception; (3) include metacommunicative signals that allow the receiver to calibrate their decoding process; and (4) accommodate the possibility that the receiver's decoding apparatus is unknown or incommensurable with the sender's encoding apparatus.

Shannon's channel coding theorem (1948) guarantees that reliable communication is possible over a noisy channel, provided the transmission rate is below the channel capacity. But Shannon's theorem assumes a fixed, known channel and a receiver that shares the sender's codebook. The absent-party problem violates both assumptions: the channel is not merely noisy but potentially reconfigured (languages change, materials decay, media become obsolete), and the receiver's codebook is unknown. This means that the standard Shannon framework must be extended in two directions: toward a theory of communication over non-stationary channels (channels whose properties change over time) and toward a theory of codebook-free communication (communication where the encoding scheme must be inferred from the message itself).

Kolmogorov complexity theory offers a partial solution: a message that contains its own decompression algorithm (a self-extracting archive, in computational terms) can in principle be decoded by any sufficiently powerful computational system, regardless of prior agreement on a codebook. The WIPP four-level message system approximates this ideal: Level 4 (technical records) provides the "codebook" for interpreting Levels 1–3. Freudenthal's Lincos attempts the same strategy for interstellar communication, building the language from mathematical primitives that are (hypothetically) universally computable. The genetic code achieves it through the ribosome — a molecular machine that is both the decoder and part of the message.

11.5 Toward a Formal Theory of Absent-Party Communication

The synthesis presented here suggests the outlines of a formal theory. An absent-party communication system can be characterized by a tuple (S, R, M, C, T, E) where S is the sender (absent), R is the receiver (unknown), M is the message (artifact), C is the channel (degrading),

T is the temporal separation, and E is the epistemic distance (degree of framework non-overlap). The system's reliability depends on the interaction of these parameters: as T increases, C degrades; as E increases, the effective codebook shrinks; as both increase together (the nuclear warning case, the SETI case), the system approaches a fundamental limit.

The key design variable is self-reference: the degree to which the message contains information about how to interpret itself. Pure self-reference (a message that is its own decoder) is the theoretical ideal; in practice, every absent-party communication system attempts to approximate it through layered encoding (WIPP), mathematical bootstrapping (Lincos), multimodal redundancy (ASPIRE ONE), or structural regularity that invites pattern-matching (the genetic code). The central open question is whether there exists a fundamental limit on self-referential encoding — a Gödelian incompleteness for communication that proves no message can fully specify its own interpretation.

11.6 Implications for Audio Production and Category Theory

For a reader versed in audio production and category theory, several connections warrant special mention. In audio production, the mastering process addresses a version of the absent-party problem: the mix engineer creates a sonic artifact for an unknown listener on unknown playback equipment in an unknown acoustic environment. The standard solution — monitoring on multiple systems, applying frequency-domain corrections that anticipate a range of playback conditions, and using reference tracks to calibrate subjective judgment — is structurally isomorphic to the WIPP strategy of multi-level, multi-modal encoding. The mix is a feedforward communication designed to survive translation across an unknown channel.

Category theory provides a potentially powerful formal language for the absent-party problem. The key concept is the functor: a structure-preserving mapping between categories. If we model the sender's conceptual framework as a category C_S (with objects as concepts and morphisms as inferential relations) and the receiver's framework as a category C_R then successful communication requires a functor $F: C_S \rightarrow C_R$ that preserves the relevant structural relations. The absent-party problem is the problem of constructing this functor when C_R is unknown. Pask's entailment mesh is, in categorical terms, a directed graph that approximates the morphism structure of C_S ; decipherment is the construction of a functor from the graph to C_R . This categorical framing has not, to our knowledge, been developed in the literature, and represents a significant opportunity for formal theory-building.

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End of Report